Introduction to Database Management Systems (DBMS)

A Database Management System (DBMS) is software designed to store, retrieve, and manage data efficiently. It serves as an interface between the database and its end-users or programs, ensuring that data is organized and easily accessible.

Key Characteristics:

- Data is stored in a structured and organized format.
- Provides security and prevents unauthorized data access.
- Supports multi-user environments with data concurrency.
- Ensures data consistency and minimizes data redundancy.

Why it matters for Web Applications:

In web development, a reliable DBMS ensures the backend can handle user logins, product inventories, and real-time transactions safely. For example, using Relational DBMS like MySQL allows developers to structure data into rows and columns, making data manipulation secure via prepared statements.